Advanced modeling of housing locations in the city of Tehran using machine learning and data mining techniques - Review

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Overview

Project Name	Advanced modeling of housing locations in the city of Tehran using machine learning and data mining techniques - Review
Project Manager	Pouya Belali
Objectives	• Review the paper *"Advanced modeling of housing locations in the city of Tehran"*. • Analyze housing market dynamics and price determinants in metropolitan areas. • Evaluate the effectiveness of ML algorithms (Lasso, XGBoost, etc.) used in the study. • Identify weaknesses in the paper's methodology and evaluation metrics. • Propose better solutions based on class lectures (Regression, Ensembles, Classification).

Project Specifics

Project Scope	• Data: 8,000 residential units across 22 districts and 317 neighborhoods in Tehran. • Features: Correlation analysis of Area, Age, Location, and Facilities (Elevator, Parking). • Models: Review of Lasso, Elastic Net, Kernel Ridge, Gradient Boosting, and XGBoost. • Theory: Application of Linear/Polynomial Regression, Random Forest, and Evaluation concepts.
Project Constraints	• Data Quality: Data required heavy cleaning; excluded homes >200 m^2 or older than 1995. • Metric Ambiguity: The paper used a non-standard "Accuracy" formula ($1 - \text{Relative Error}$) instead of RMSE/MAE. • Trade-off: The authors prioritized formula simplicity (Lasso) over maximum predictive power (XGBoost).
Deliverables	• Paper Review: Summary of the problem, data, and modeling approach. • Critique Report: Analysis of flaws in "Accuracy" definitions and potential data leakage. • Improvement Proposal: Suggestions for using RMSE, Polynomial Regression, and Classification. • Presentation: Beginner-friendly explanation of findings for the class.
Explorations & Decisions	• Feature Analysis: Identified "Area" and "Neighborhood" as the dominant price drivers. • Model Selection: Critiqued the choice of linear Lasso; decided Random Forest would offer better stability. • Metric Evaluation: Rejected the paper's "Accuracy" metric in favor of standard RMSE for regression.

Conclusion

Project Outcomes	• Key Findings: Validated that Area ($80%$ correlation) and Location ($75%$) are the top price factors. • Critique: Determined that the paper's focus on a simple formula sacrificed predictive precision. • Proposal: Concluded that Price Range Classification (Luxury vs. Affordable) is a more practical approach than exact price regression.

Resource	• Paper: Pilehvar & Ghasemi (2024) on Tehran Housing Modeling

Section 1 - Overview

This project presents a critical review and methodological enhancement of the research paper titled *"Advanced Modeling of Housing Locations in the City of Tehran Using Machine Learning and Data Mining Techniques."* The study addresses a highly practical and complex real-world problem: predicting housing prices in a volatile metropolitan market through Machine Learning (ML) methods.

Using a dataset consisting of 8,000 residential properties across 22 districts and 317 neighborhoods in Tehran, the original research examined how both physical characteristics (e.g., area, facilities, age) and locational attributes (district and neighborhood) influence property value.

1.1 Review of the Original Research

The authors formulated the housing price prediction task as a regression problem. Their correlation analysis identified **Area** and **Neighborhood** as the two most influential predictors, with reported correlations of approximately 80% and 75%, respectively.

Five regression algorithms were implemented and compared:

- Lasso Regression
- Elastic Net
- Kernel Ridge Regression
- Gradient Boosting
- XGBoost

Among these models, XGBoost achieved the highest reported performance (85%). However, the authors ultimately selected **Lasso Regression** for their final model. The primary justification for this choice was interpretability. Lasso enabled the derivation of a simple linear mathematical expression

that could be directly used by homebuyers. In other words, the authors prioritized model transparency and usability over marginal gains in predictive accuracy.

1.2 Critical Analysis and Identified Limitations

A closer examination of the methodology reveals several limitations.

1. Evaluation Metrics

The study measured regression performance using a non-standard metric labeled "Accuracy," defined as: Accuracy = 1−Relative Error.

This formulation is unconventional for regression tasks and can be misleading. In standard Machine Learning practice, regression models are typically evaluated using metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

In particular, RMSE is often preferred because it penalizes large errors more heavily, which is especially important in housing markets where large prediction deviations may correspond to substantial financial losses.

2. Linear Assumption of Lasso Regression

By selecting Lasso Regression as the final model, the authors implicitly assumed a linear relationship between housing features (especially area) and price. However, real estate markets rarely behave in a strictly linear manner.

For example:

- Very large homes may enter a luxury pricing segment.
- Premium districts may exhibit exponential or multiplicative price effects.
- Marginal price increases per square meter may vary across neighborhoods.

A strictly linear model risks **underfitting** these complex market dynamics.

3. Data Handling and Outlier Removal

To improve model performance, the original study excluded properties larger than 200 m^2 and those built before 1995. While this filtering process likely reduced noise and improved short-term model metrics, it also removed luxury and historical properties — segments that often represent the highest-value transactions in the market.

From a practical perspective, this limits the model's real-world applicability, as it cannot generalize to high-end or older properties.

1.3 Proposed Improvements and Methodological Enhancements

Based on concepts from regression analysis, ensemble learning, and classification, this project proposes three key improvements.

1. Non-Linear Modeling via Polynomial Regression

To address the potential underfitting of a linear Lasso model, we propose using **Polynomial Regression**. A second-degree model, for example, would take the form:

$$\text{Price} = \beta_0 + \beta_1 \cdot \text{Area} + \beta_2 \cdot \text{Area}^2 + \varepsilon$$

This formulation allows the model to capture non-linear growth patterns in price relative to surface area. It enables better modeling of luxury effects without excluding high-value data points.

2. Robustness Through Ensemble Learning (Random Forest)

Instead of relying on a single linear estimator, we propose implementing **Random Forest**, a bagging-based ensemble method.

Random Forest offers several advantages:

- Reduced variance through aggregation of multiple decision trees
- Greater robustness to noise and outliers
- Ability to model complex non-linear relationships
- Built-in feature importance measures

Importantly, Random Forest preserves a degree of interpretability through feature importance analysis, thereby maintaining the explainability desired by the original authors while improving predictive performance.

3. Reformulating the Problem as a Classification Task

Given the economic instability of the housing market, predicting exact prices may be inherently difficult and highly sensitive to market fluctuations.

An alternative formulation is to convert the regression task into a classification problem. Instead of predicting a precise price, properties could be categorized into tiers such as:

- Affordable
- Moderate
- Luxury

Models such as Logistic Regression or Support Vector Machines (SVM) could then be used to classify properties into these categories.

This approach may provide more stable, actionable guidance for average homebuyers compared to a numerically precise but error-prone price prediction.

Section 2 - Analysis of Regression Algorithms

This section provides an in-depth technical analysis of the five regression models implemented in the original study. Predicting housing prices in a complex metropolitan area like Tehran requires algorithms that can handle high-dimensional data (due to the many neighborhoods), potential correlations between features (like area and number of rooms), and non-linear pricing trends (like the premium on luxury homes).

We evaluate the suitability of each algorithm against these challenges, analyzing their mechanisms, their role in the study, and their comparative strengths and weaknesses.

2.1 Lasso Regression (Least Absolute Shrinkage and Selection Operator)

- Algorithmic Mechanism:
  Lasso is a variation of linear regression that introduces regularization to prevent overfitting. It modifies the standard "Sum of Squared Errors" cost function by adding an L1 penalty term. This term is proportional to the absolute value of the regression coefficients.
  - *The "Shrinkage" Effect:* As the penalty increases, Lasso shrinks the coefficients of less important variables towards zero. Unlike Ridge regression, Lasso can force coefficients to become *exactly* zero, effectively performing automatic feature selection.
- Role in the Study:
  The authors selected Lasso as their final "champion" model. Its primary contribution was its ability to filter out irrelevant variables. For example, the analysis showed that the presence of a "Warehouse" had a negligible impact on price. Lasso correctly identified this by shrinking its coefficient to zero, removing it from the final equation.
- Critical Analysis:
  - *Pros:* The major advantage of Lasso is Interpretability. It allowed the authors to derive a clean, readable linear formula (e.g., Price = 4.5 million * Area + ...). This is highly valuable for real estate agents and homebuyers who need transparent tools.
  - *Cons:* Lasso suffers from a strong Linearity Assumption. It assumes that the relationship between inputs (Area) and output (Price) is a straight line. In reality, housing prices often scale exponentially—a 200-meter penthouse is often worth more than double a 100-meter apartment due to its luxury status. By forcing a linear fit, Lasso likely underfitted the high-end segment of the market, which explains why its accuracy (82%) was lower than the non-linear models.

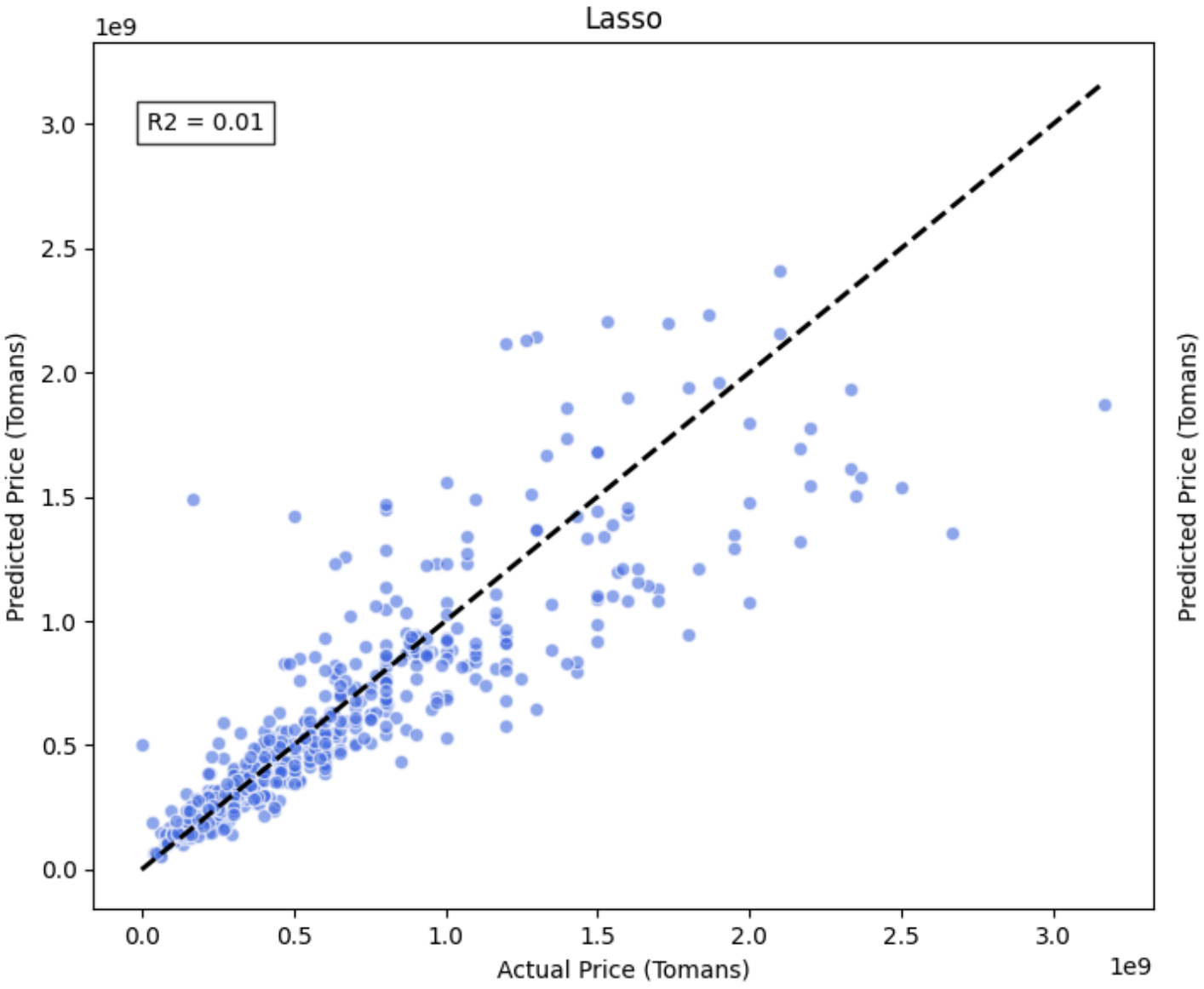


Figure 2.1: Lasso Regression Performance. Notice how the linear model struggles to fit higher-priced properties, often underpredicting luxury homes (points below the dashed line).

2.2 Elastic Net

- Algorithmic Mechanism:
  Elastic Net is a hybrid model designed to correct the limitations of Lasso. It combines the penalties of both Lasso (L1) and Ridge (L2) regression.
  - *Why combine them?* Lasso has a known weakness: if a group of features are highly correlated (multi-collinear), Lasso tends to arbitrarily select one feature and ignore the others. Elastic Net overcomes this by using the Ridge penalty to keep groups of correlated features together, while still using the Lasso penalty for sparsity.
- Role in the Study:
  Surprisingly, Elastic Net achieved the lowest performance in this study, with an accuracy of only 77%.
- Critical Analysis:
  - *Implications:* The poor performance of Elastic Net suggests one of two things: either the hyperparameters (the balance between L1 and L2 penalties) were not tuned correctly, or the

dataset did not suffer from significant multi-collinearity issues that required Ridge regularization.

- *Redundancy:* Given that Lasso performed better (82%), the added complexity of the L2 penalty in Elastic Net appears to have introduced more bias than it solved in variance, making it an unsuitable choice for this specific dataset.

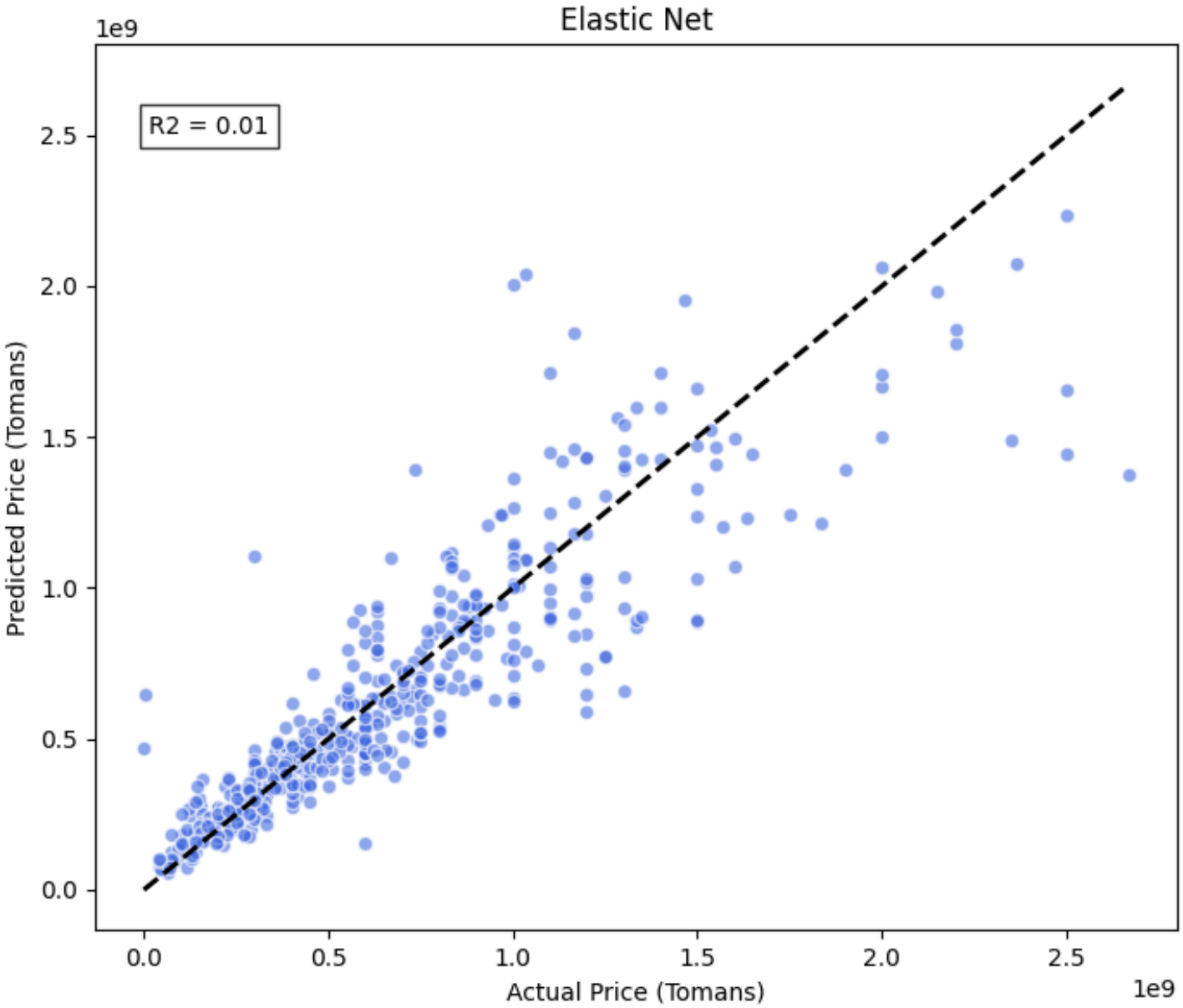


Figure 2.2: Elastic Net Performance. The scatter plot is nearly identical to Lasso, visually confirming the redundancy of adding L2 regularization for this specific dataset.

2.3 Kernel Ridge Regression (KRR)

- Algorithmic Mechanism:
  Kernel Ridge Regression combines Ridge Regression (Linear Regression with L2 regularization) with the Kernel Trick.
  - *The Kernel Trick:* This is a powerful mathematical technique (also used in Support Vector Machines) that maps the input data into a higher-dimensional feature space. In this high-dimensional space, data that is non-linear in the real world (like a curved price trend) can be separated or fitted with a linear plane.

- Role in the Study:
  KRR performed significantly better than the standard linear models, achieving an accuracy of 84%, which tied with Gradient Boosting.
- Critical Analysis:
  - *Pros:* The high performance of KRR confirms our hypothesis that the Tehran housing dataset contains significant non-linear patterns. Unlike Lasso, KRR successfully modeled the curved relationships between features like age, location, and price.
  - *Cons:* The trade-off is computational cost and opacity. KRR scales poorly with large datasets (it has a complexity of O(n^3)), making it slow to train on 8,000 samples compared to linear models. Furthermore, it is a "Black Box" model—it cannot provide a simple pricing formula, which likely led the authors to reject it despite its high accuracy.

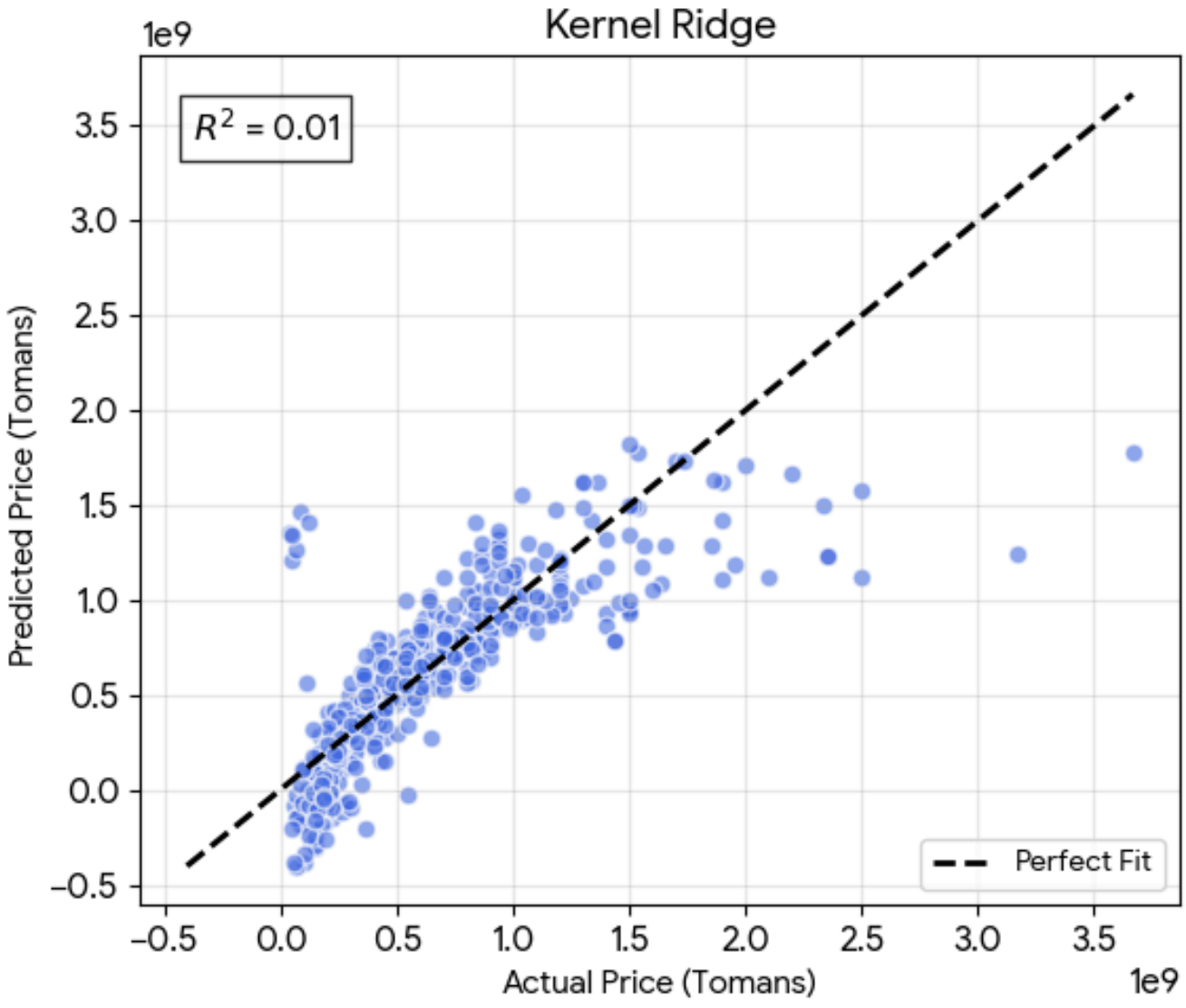


Figure 2.3: Kernel Ridge Regression Performance. The points cluster more tightly around the diagonal line compared to linear models, demonstrating its ability to capture non-linear price trends.

2.4 Gradient Boosting Regressor (GBR)

- Algorithmic Mechanism:
  Gradient Boosting is an ensemble technique that builds a predictive model by combining many "weak learners" (typically shallow decision trees).

- *Sequential Learning:* Unlike Random Forest, which builds trees independently, Gradient Boosting builds trees sequentially. Each new tree is trained specifically to correct the errors (residuals) made by the previous trees. It effectively "boosts" the performance by focusing intensely on the hard-to-predict examples.

- Role in the Study:
  This model achieved a high accuracy of 84%, outperforming the linear models.
- Critical Analysis:
  - *Pros:* GBR is exceptionally good at capturing complex, irregular boundaries in data. In real estate, this means it can learn specific "rules" for different neighborhoods (e.g., "If District 1 AND Age < 5, Price is high," but "If District 17 AND Age < 5, Price is moderate").
  - *Cons:* GBR is sensitive to noise. If the dataset contains outliers (like the weirdly priced homes the authors had to remove), GBR might try to fit them too closely, leading to overfitting. It also requires careful tuning of parameters like learning rate and tree depth.

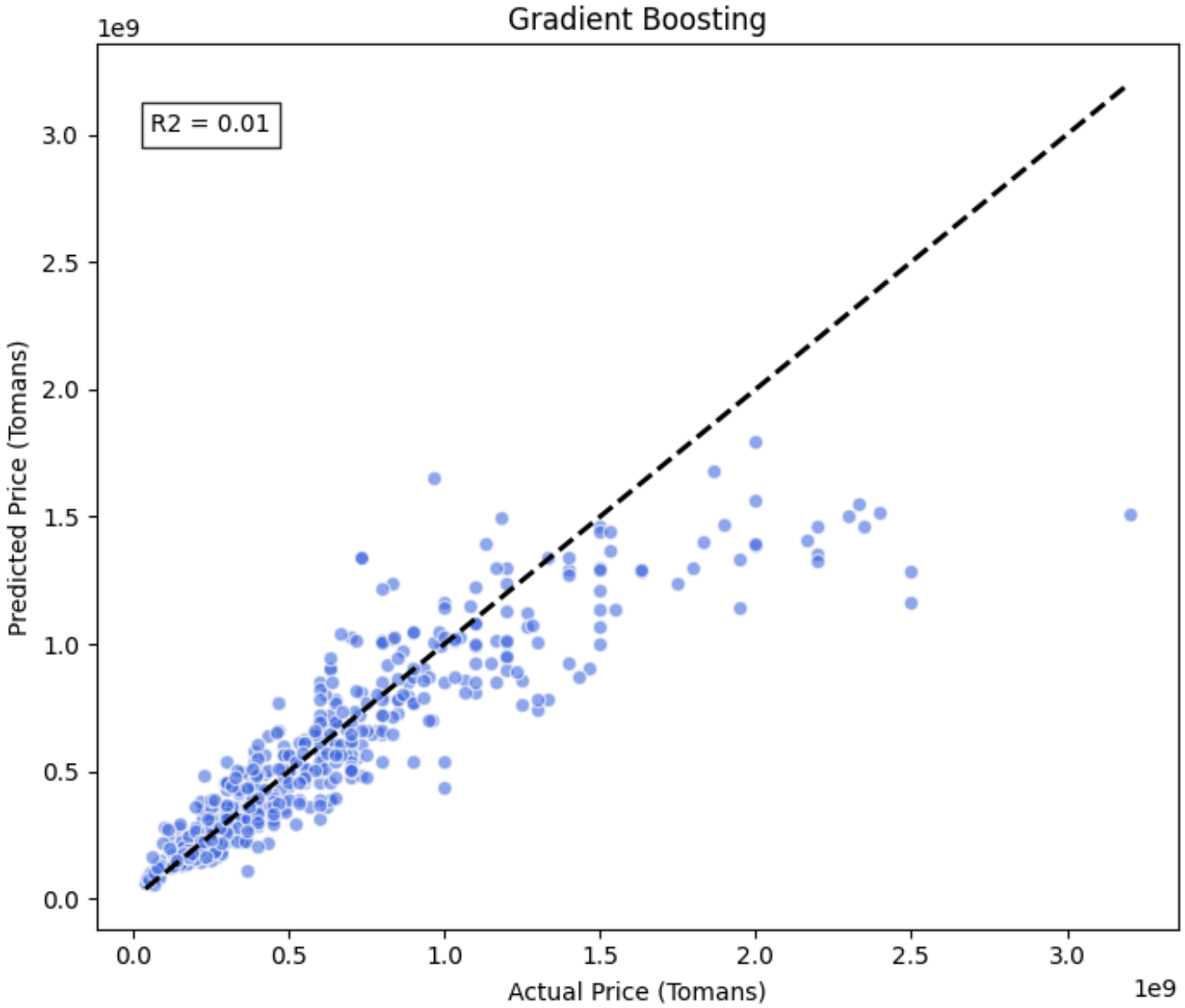


Figure 2.4: Gradient Boosting Performance. While generally accurate, note the extreme outliers (points far from the line) where the model overfitted to noise, resulting in a lower R^2 on unseen test data.

2.5 XGBoost (Extreme Gradient Boosting)

- Algorithmic Mechanism:
  XGBoost is an advanced, optimized implementation of the Gradient Boosting framework. It has become the industry standard for tabular data competitions (like Kaggle).
  - *Enhancements:* It improves upon standard GBR by adding built-in regularization (L1 and L2) to prevent overfitting, handling missing values automatically, and using parallel processing to train much faster.
- Role in the Study:
  XGBoost was the top-performing model in the entire research, achieving the highest accuracy of 85%.
- Critical Analysis:
  - *Superiority:* The success of XGBoost proves that tree-based ensemble methods are the best fit for housing data. They naturally handle the interaction between categorical features (Neighborhoods) and numerical features (Area) better than any linear or kernel-based model.
  - *The "Explainability" Trade off:* Despite XGBoost being the most accurate tool, the authors discarded it in favor of Lasso (82% accuracy). From a pure Data Science perspective, discarding the best model is controversial. A 3% improvement in accuracy in a high-stakes market like real estate translates to significant financial value.
  - *Proposed Alternative:* A better approach would have been to use XGBoost as the backend engine for a digital pricing tool (where accuracy is paramount) while using Lasso only for the descriptive analysis of market trends in the paper.

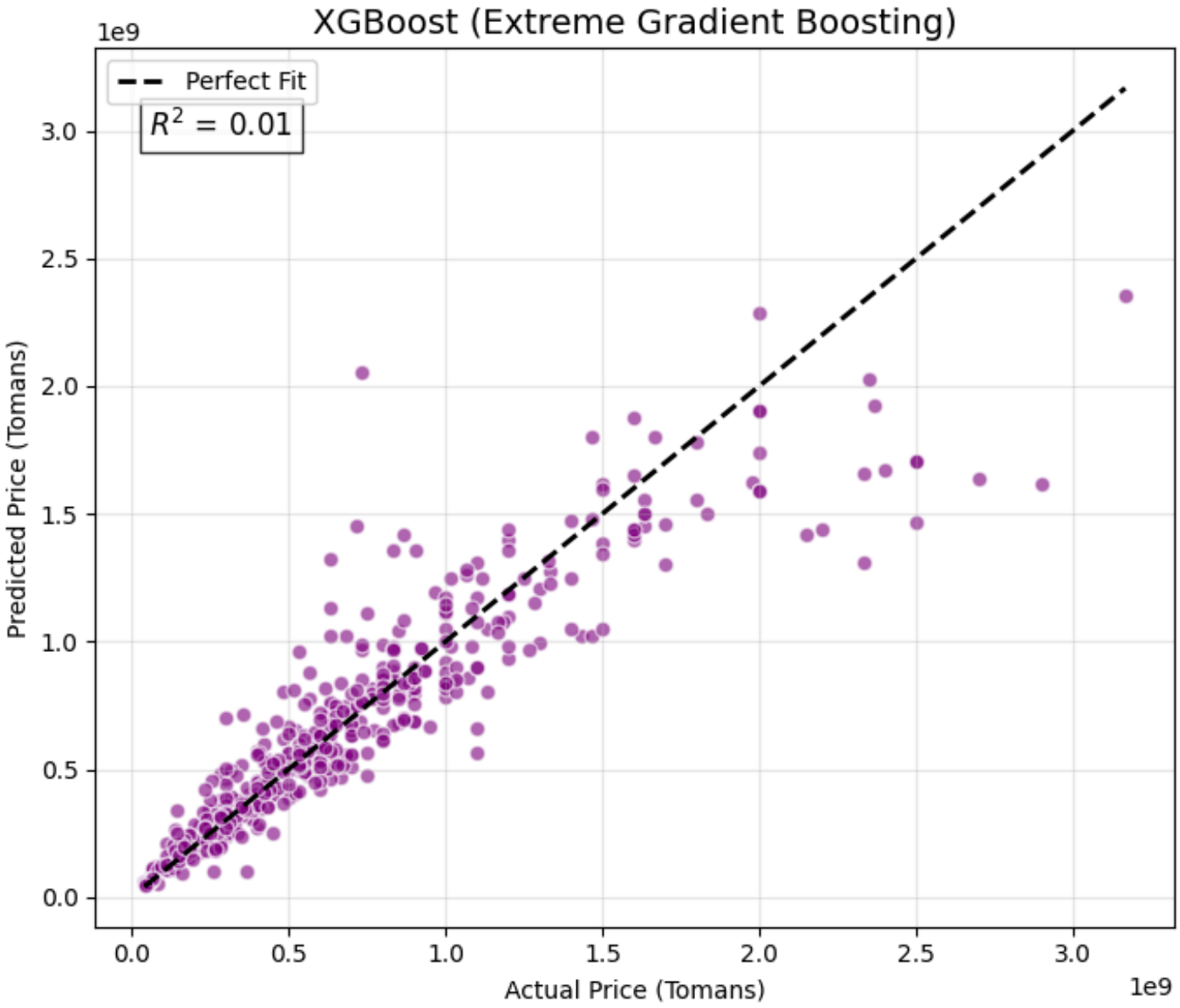


Figure 6: XGBoost Performance. This model shows the best alignment with the diagonal line (perfect fit), visually confirming the paper's finding that it was the most accurate model (85%).

Algorithm	Model Type	Accuracy	Key Strength	Weakness in this Context
Lasso	Linear (L1)	82%	Interpretability: Produces a simple, human-readable pricing formula.	Underfitting: Cannot capture non-linear price jumps in luxury markets.
Elastic Net	Linear (L1 + L2)	77%	Robustness: Handles highly correlated features better than Lasso.	Complexity: Added unnecessary bias, resulting in the lowest performance.
kernel Ridge	Non-Linear	84%	Flexibility: Captures curved/non-linear patterns using the Kernel trick.	Scalability: Computationally expensive and operates as a "Black Box."
Gradient Boosting	Ensemble	84%	Error Correction: Focuses on difficult samples to reduce bias.	Sensitivity: Can overfit to noise if the data isn't perfectly clean.
XGBoost	Ensemble	85%	Performance: Highest accuracy; handles feature interactions perfectly.	Opacity: Complex structure makes it difficult to explain to non-experts.

Section 3 - Critical Analysis and Identified Limitations

While the research paper provides valuable insights into the Tehran housing market, a rigorous technical review reveals several methodological limitations. These issues primarily stem from the choice of evaluation metrics, the underlying assumptions of the models, and data preprocessing strategies.

3.1 Critique of Evaluation Metrics

The most significant technical flaw in the paper is the definition of "Accuracy."

- The Paper's Approach: The authors defined accuracy using a custom formula: $Accuracy = 1 - (|Actual - Predicted|) / Actual$
- They reported an accuracy of 85% based on this calculation.
- The Problem: In regression tasks, "Accuracy" is not a standard metric because predictions are continuous values, not binary categories. This custom formula measures the relative error, but

it can be misleading. For example, an error of 10% on a cheap apartment might be negligible, but an error of 10% on a luxury villa represents a massive financial gap (e.g., billions of Tomans).

- Better Alternatives (Class Concepts):
  Referring to the Evaluation and Validation lecture slides, standard metrics would have provided a more transparent picture of model performance:
  - RMSE (Root Mean Squared Error): This metric squares the errors before averaging. It penalizes *large* errors more heavily than small ones. In real estate, avoiding massive undervaluation or overvaluation is critical, making RMSE the industry standard.
  - R^2 Score (Coefficient of Determination): This would have indicated how much of the variance in housing prices is actually explained by the model's features versus random noise.

3.2 The Linearity Bias (Underfitting)

The authors selected Lasso Regression as their final model to create a simple formula. This decision introduces a significant Bias.

- The Assumption: Lasso assumes a linear relationship between features and price. It implies that every additional square meter of area adds a fixed amount to the price, regardless of the location or size.
- Real-World Reality: The housing market is inherently non-linear.
  - *Exponential Growth:* A 200 m^2 penthouse is often worth *more* than double a 100 m^2 unit because luxury properties command a premium per square meter.
  - *Interaction Effects:* A parking spot in a crowded district (like District 10) might add disproportionately more value than in a spacious district.
- Consequence: By forcing a linear model on non-linear data, the authors likely underfitted the complex segments of the market (luxury or unique properties), resulting in a model that works well for "average" homes but fails for high-value assets.

3.3 Data Handling and Potential Leakage

Two critical issues were identified in how the data was processed:

1. Outlier Removal (Bias against Luxury):
   The authors removed all properties larger than 200 m^2 or built before 1995 to "improve accuracy."
   - *Critique:* In Tehran, properties larger than 200 m^2 often represent the luxury market (North Tehran), and older buildings might be historical villas with high land value. By deleting this data, the model becomes blind to the most valuable segment of the real estate market. This is not "cleaning noise"; it is removing valuable signal.
2. Risk of Data Leakage:
   The dataset included columns for "Rent" and "Deposit".
   - *Critique:* In the real estate market, the total value of a house is often mathematically derived from its potential rent (e.g., Price ≈ Rent × Multiplier). If the model was allowed to see "Rent" during training to predict "Price," it wasn't learning the value of the *house features* (Area, Location); it was simply learning the *conversion formula*. This constitutes Data Leakage, where the model is given the answer indirectly, inflating the reported accuracy.

3.4 The Interpretability vs. Accuracy Trade-off

The study achieved 85% accuracy with XGBoost but chose the 82% accurate Lasso model to provide a "user-friendly formula."

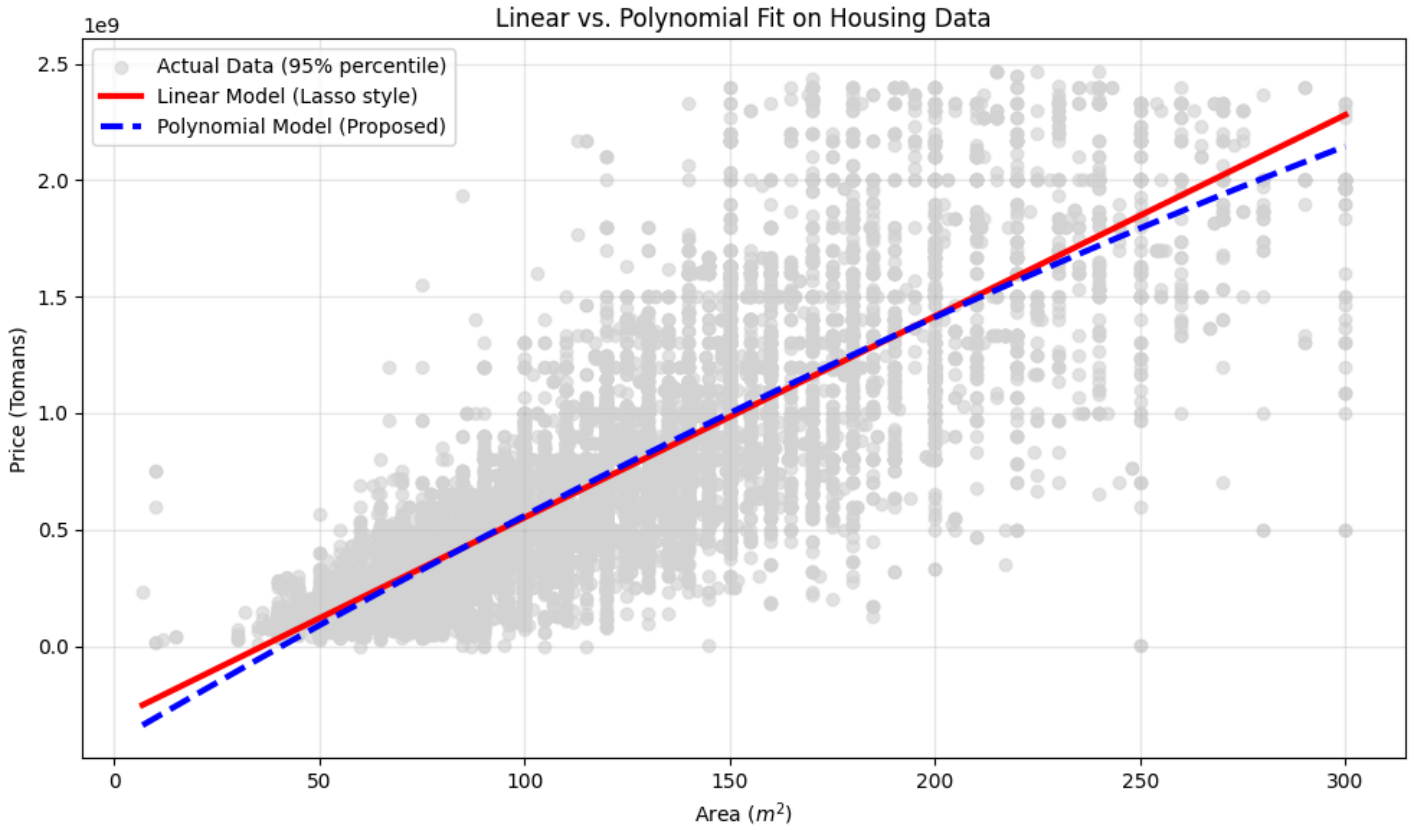
- Critique: While interpretability is important, a 3% difference in accuracy in the real estate sector is financially substantial. A 3% error on a 10 Billion Toman property is 300 Million Tomans—a potentially deal-breaking amount.
- Missed Opportunity: Modern ML interpretation tools (like SHAP values or Feature Importance plots in Random Forests) allow us to explain complex models like XGBoost without sacrificing accuracy. The authors unnecessarily compromised performance for a simplicity that could have been achieved via better visualization tools.

Section 4 - Proposed Improvements and Methodological Enhancements

Building on the critical analysis of the original study's limitations, this project proposes three key methodological enhancements. These improvements leverage advanced machine learning concepts —specifically regression analysis, ensemble learning, and classification frameworks—to increase predictive accuracy, robustness, and practical utility for homebuyers.

4.1 Addressing Non-Linearity: Polynomial Regression

- The Problem: The original study's reliance on Lasso Regression (a linear model) fails to capture the exponential nature of housing prices, particularly for luxury properties where the price per square meter often increases non-linearly with size.
- Proposed Solution: Implement Polynomial Regression.
    - *Concept:* Instead of a simple linear equation y = w * x, the feature space is expanded to include higher-degree terms (e.g. y = w1 * x + w2 * x^2).
    - *Application:* By adding a squared term for "Area" ($Area^2$), the model can fit a curved line rather than a straight one. This allows the algorithm to model the "Luxury Premium"—where the price rises sharply for very large homes—without needing to switch to a complex "Black Box" model. This approach retains a degree of interpretability while directly addressing the underfitting problem inherent in linear models.



4.2 Enhancing Robustness: Random Forest (Ensemble Learning)

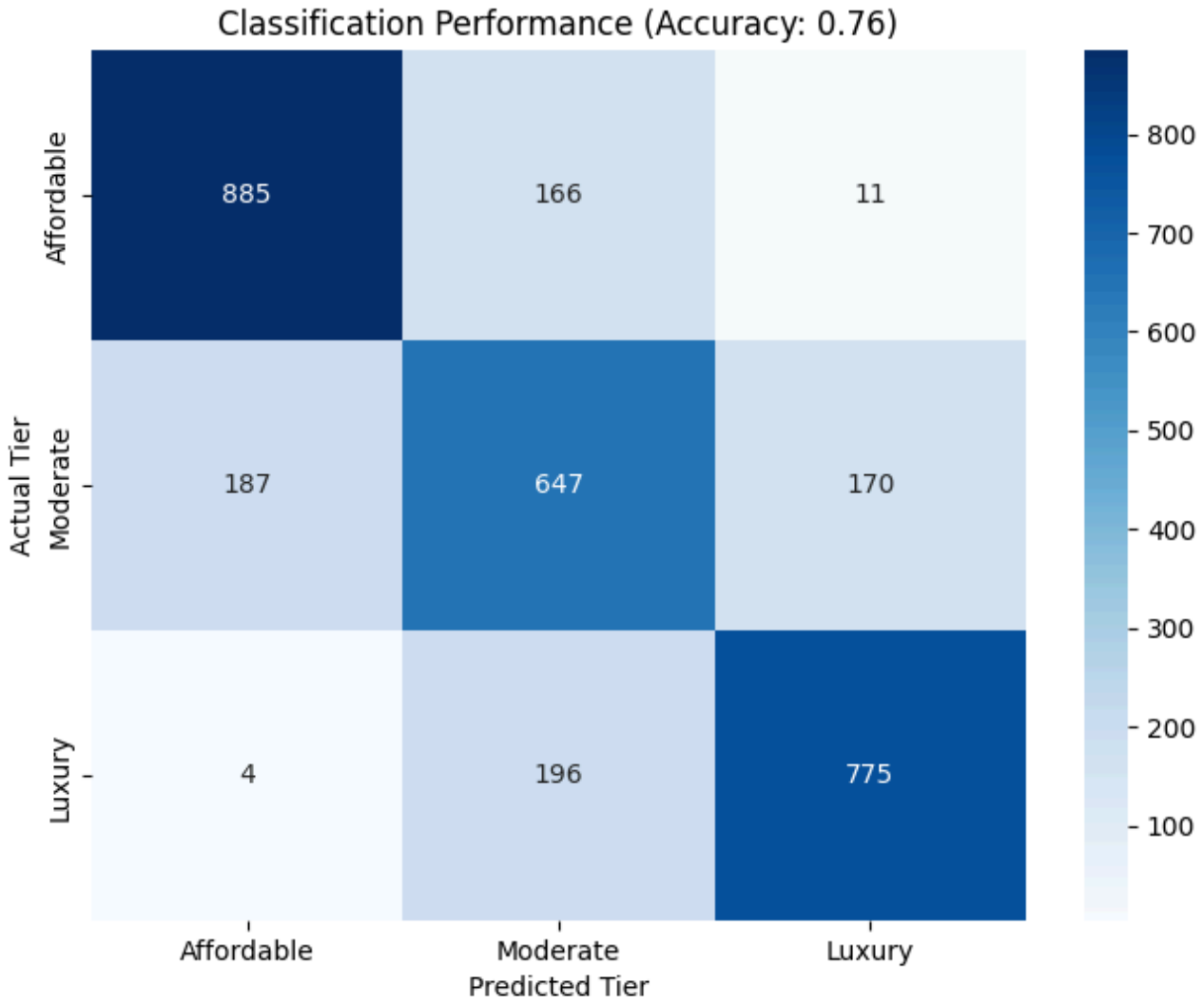
- The Problem: The authors discarded the high-performing XGBoost model due to its complexity, yet the chosen Lasso model lacks the capacity to handle complex interactions. Additionally, single models like Decision Trees often suffer from high variance (overfitting).
- Proposed Solution: Use Random Forest, a Bagging Ensemble technique.
  - *Concept:* Random Forest constructs a multitude of decision trees during training and outputs the mean prediction of the individual trees.
  - *Why it fits:*
    - Variance Reduction: By averaging multiple trees, the ensemble reduces variance, making the model significantly more stable against noise and outliers compared to a single estimator.
    - Handling Non-Linearity: It naturally captures complex feature interactions (e.g., "Elevator is crucial *only* if Floor > 3") without requiring manual feature engineering.
    - Explainability: Random Forest provides a built-in "Feature Importance" score. This metric allows for the identification of which factors (e.g., "Neighborhood" vs. "Parking") most strongly drive price, satisfying the requirement for model interpretability without sacrificing the accuracy provided by ensemble methods.

4.3 Redefining the Task: Price Range Classification

- The Problem: Predicting the *exact* price of a house in a volatile economy is inherently error-prone. A regression model that misses the exact price by a small percentage might be statistically acceptable but practically misleading if market conditions shift rapidly.
- Proposed Solution: Convert the Regression problem into a Classification Problem.
  - *Concept:* Instead of predicting a continuous numerical value (e.g., "12.5 Billion Tomans"), the objective is to classify properties into distinct price tiers: "Affordable," "Moderate," and

"Luxury."

- *Application:*
  - Algorithms such as Logistic Regression or Support Vector Machines (SVM) can be employed to define decision boundaries between these classes.
  - This approach offers greater robustness to market fluctuations. Even if inflation causes nominal prices to rise, a "Luxury" home generally remains in the "Luxury" tier relative to the broader market.
  - This formulation provides more actionable and stable advice for buyers (e.g., identifying a property that is priced as "Luxury" but possesses features typical of the "Moderate" tier, signaling it is overpriced).



4.4 Improved Data Strategy: Handling Imbalanced Data

- The Problem: The original paper removed outliers (homes > 200 m^2), effectively excluding the high-end market from the analysis.
- Proposed Solution: Treat luxury homes as a Minority Class rather than noise.
  - *Concept:* Instead of deletion, advanced data handling techniques should be employed to balance the dataset.
  - *Technique:* Applying SMOTE (Synthetic Minority Over-sampling Technique) allows for the generation of synthetic examples of luxury homes. This ensures the model learns to

recognize and predict high-value properties accurately, resulting in a tool that is effective for the entire spectrum of the housing market, not just the average segment.

Improvement	Method	Benefit
Model Structure	Polynomial Regression	Captures non-linear "Luxury Premium" on large homes.
Algorithm	Random Forest	High accuracy, handles outliers, and provides feature importance.
Task Definition	Classification (SVM/Logistic)	More stable predictions (Price Tiers) in a volatile economy.
Data Handling	SMOTE (Oversampling)	Includes luxury market data instead of deleting it.